MSE analysis for IOTC albacore

Current status and future steps

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IOTC WPM MSE

ABC OMs

Biology

- Age, sex and quarterly structured population model
- Timeline: 2000 to 2020 (covers all existing cohorts)
- Beverton-Holt with exploited equilibrium initialisation
- Reproduces stock status variables: SB_{MSY}, H_{MSY}, depletion to SB_0

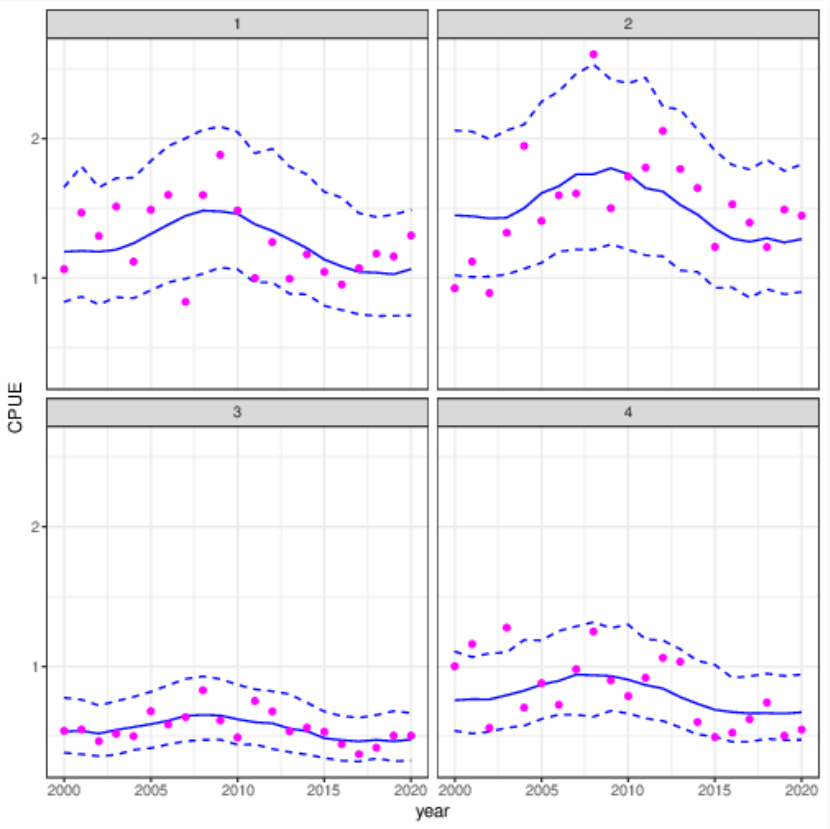
Fisheries

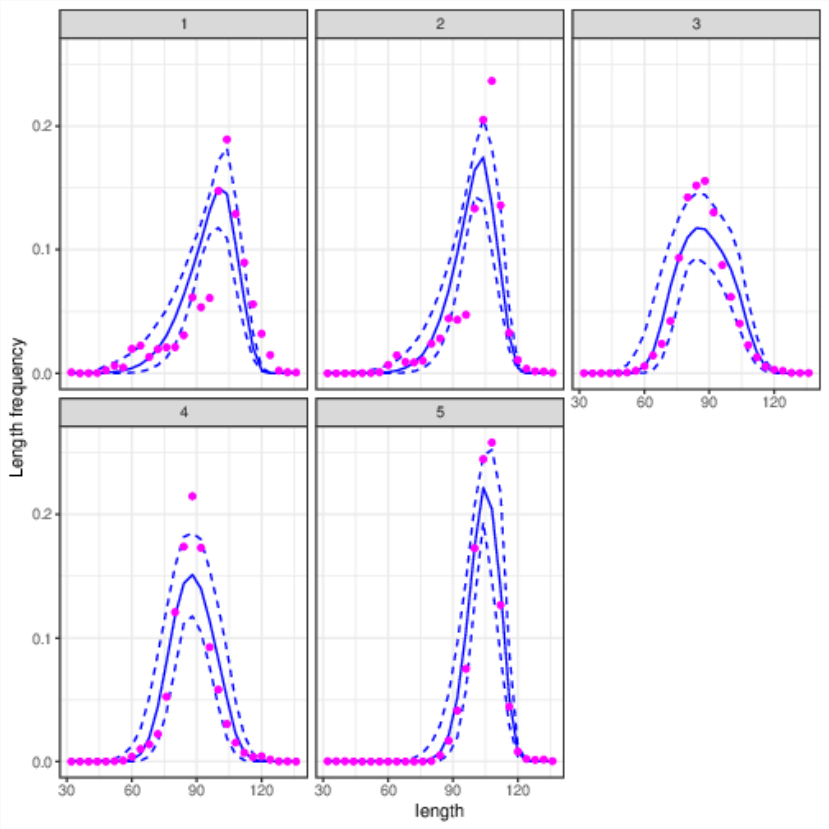
- Merges "common" seasonal LL fleets 1–4
- Retains single PS and "Other" fleet: 6 in total
- Size data from LL and PS data (aggregated across time)
- LL CPUE from a given fleet / area

ABC uncertainty

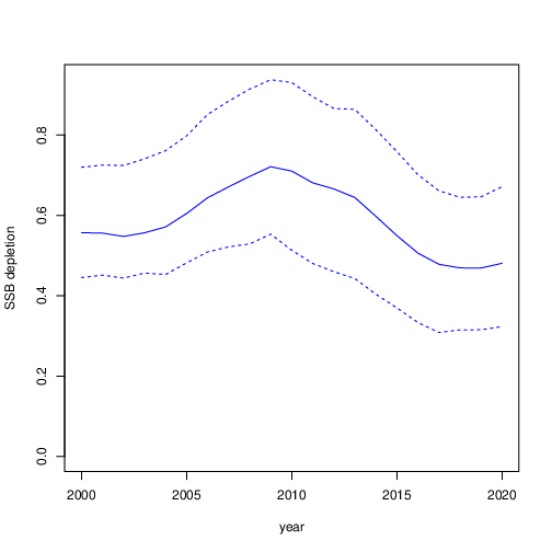
- Steepness & M: covariance joint prior (not discrete grid)
- σ_r^2: (i) fixed at 0.3; (ii) estimated with prior CI 0.2–0.5
- LF: weight/influence (aggregating and ABC discrepancy)
- LL catchability: alternative 1% annual increasing trend
- CPUE series: seasonal q using fleet 1 and 3 *separately*

OM - Fits to CPUE series & LF

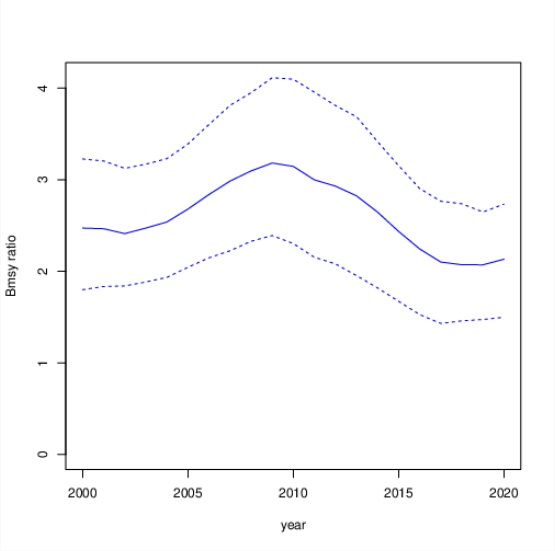




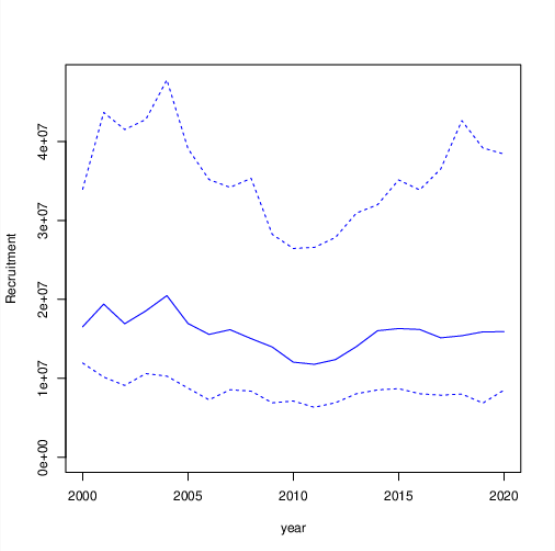
OM - Population dynamics



Depletion

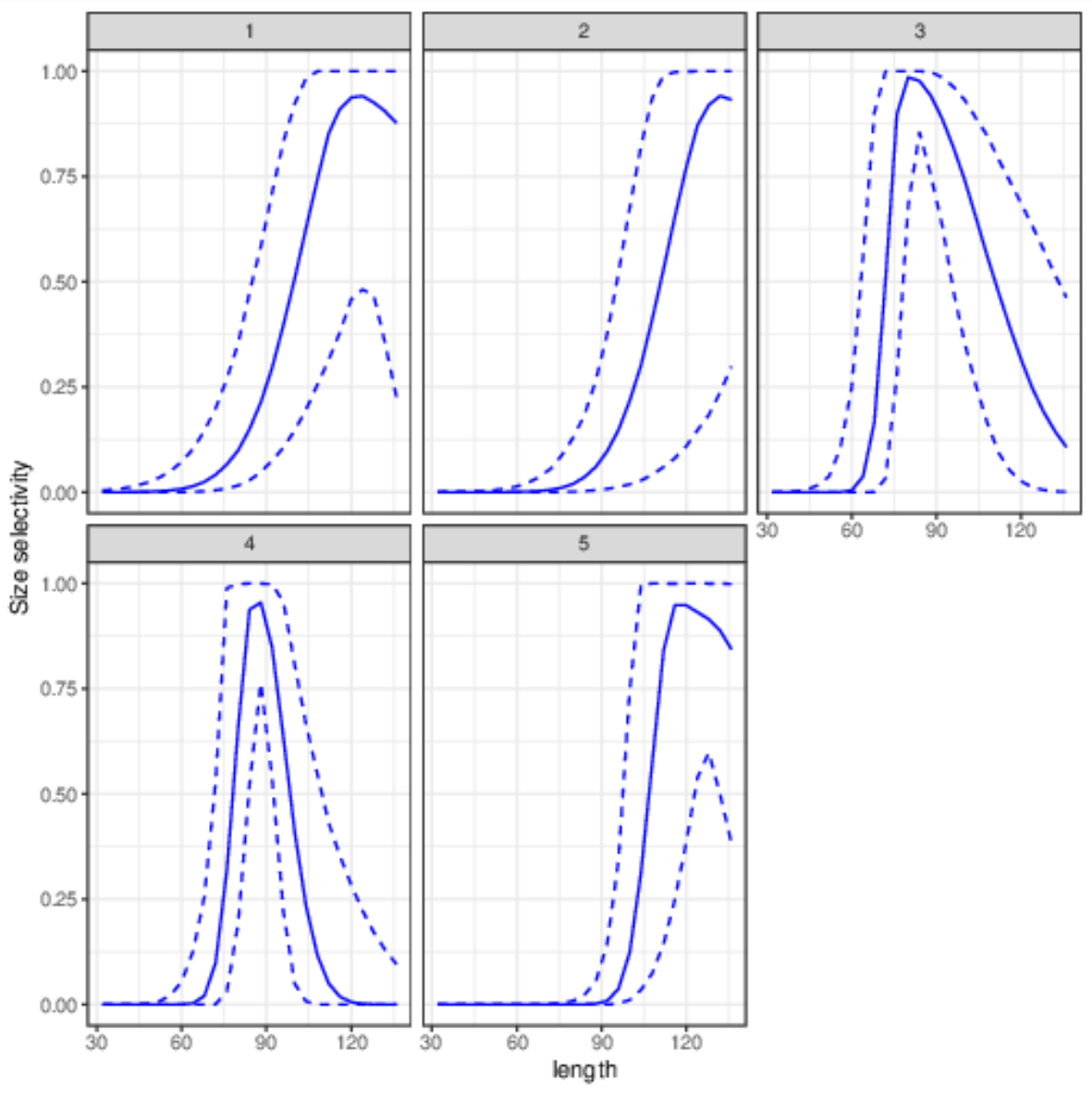


B/B_{MSY}



Recruitment

OM - Size selectivity



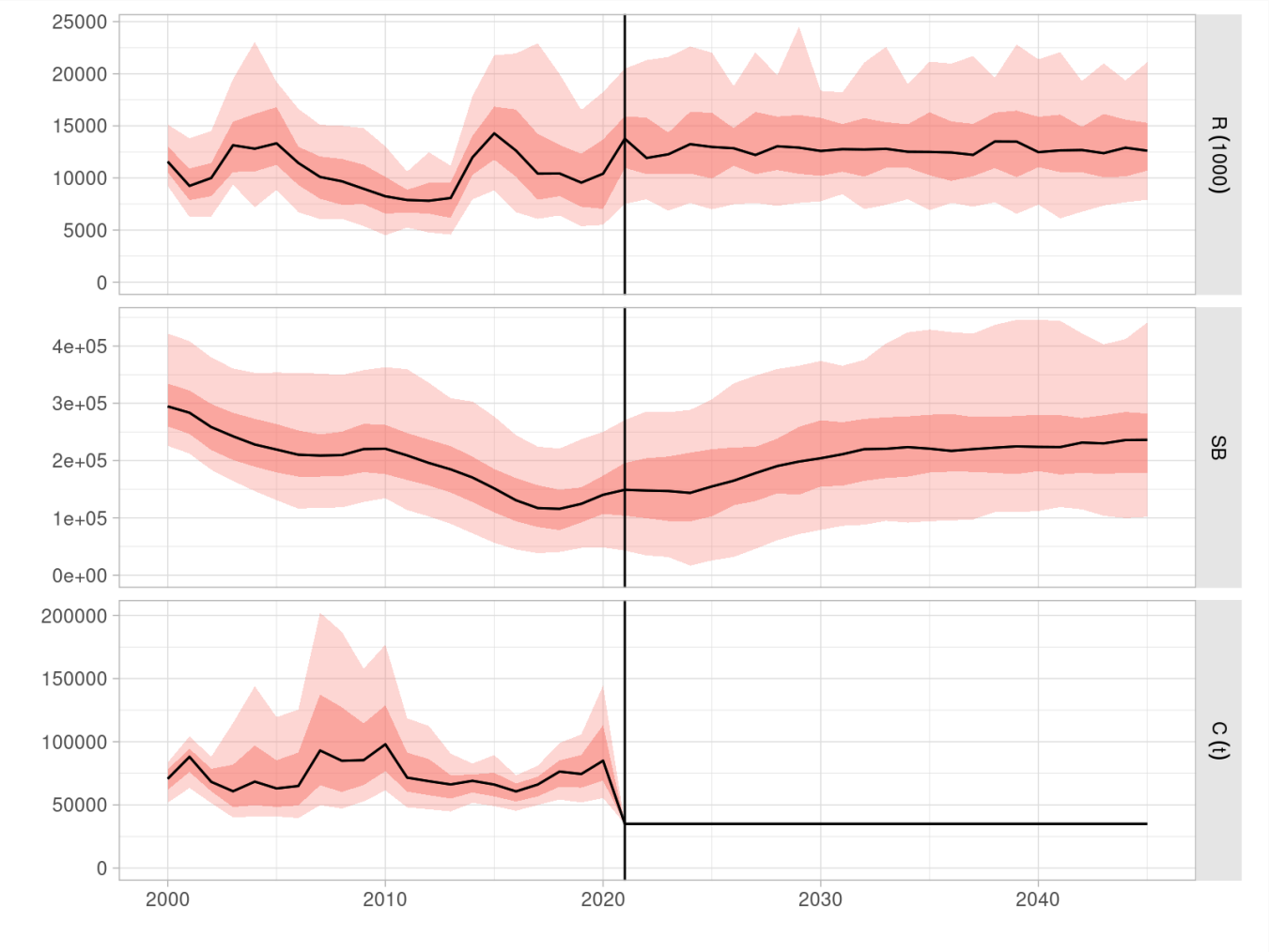
OM grid

- **R1** CPUE fleet 1, SSB but *not* H_{MSY} priors
  - CPUE fleet 3 (**R1a**)
  - Overfishing penalty (**R1b**)
- **R2**, R1 + estimated σ_r^2
  - CPUE fleet 3 (**R2a**MSY)
  - Overfishing penalty (**R2b**)
- **R3** R1 + 1% effort creep
  - CPUE fleet 3 (**R3a**)
  - Overfishing penalty (**R3b**)

Robustness OMs (WPM15

23. The WPM **NOTED** that the reference OM is defined as the model that includes LL CPUE 1, with SSB priors, recruitment variability, and the overfishing penalty (Model R2b). Robustness OMs include using LL CPUE 3 in place of LL CPUE 1 (Model R2a) and adding 1% effort creep (Models R3b). The WPM **AGREED** to the reference OM but **SUGGESTED** to use CPUE 1 instead of the CPUE 3 in the two robustness OMs suggested.

OM projections - C = 35,000 t



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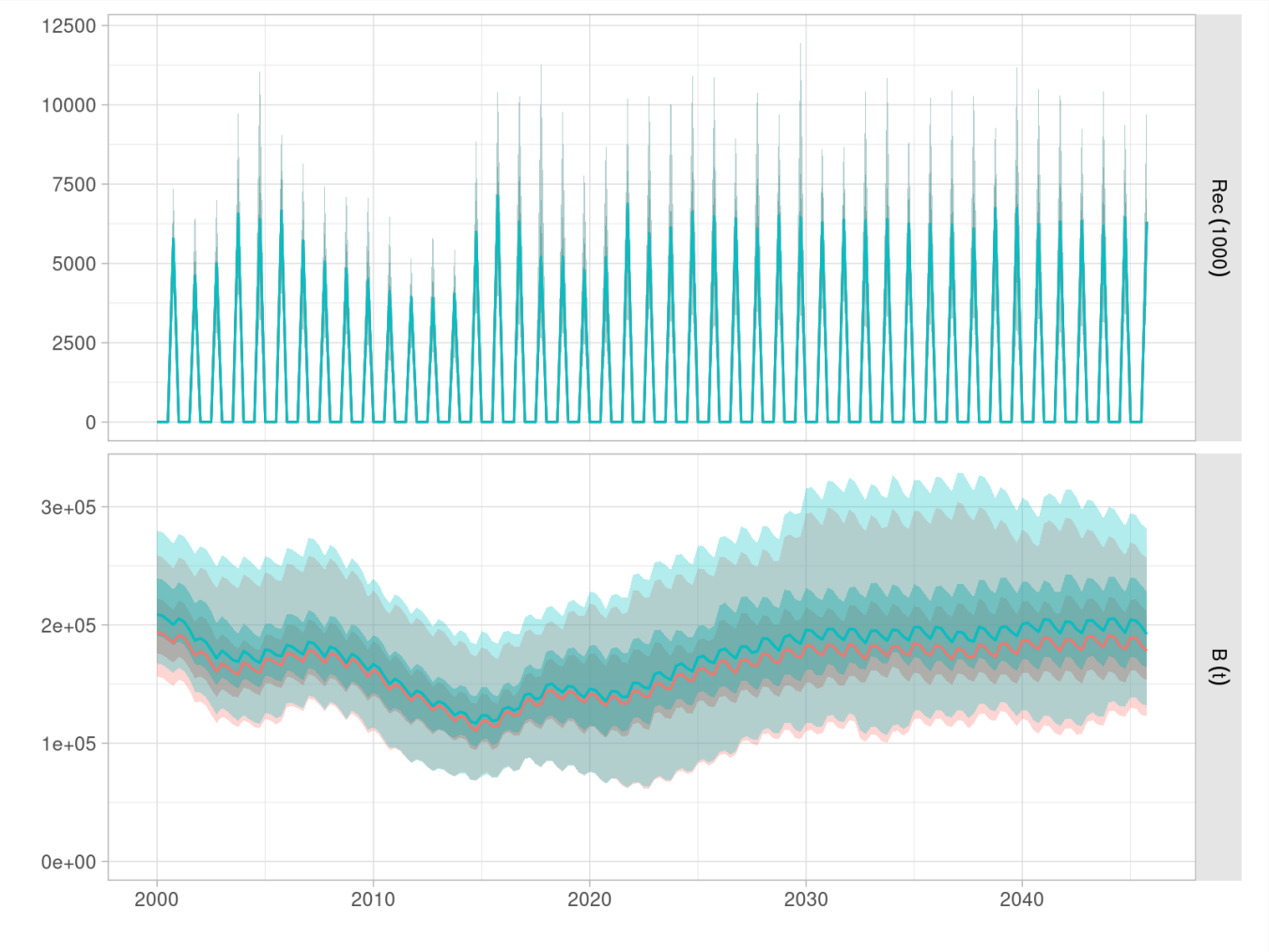

Reference points

- SB/SB_{MSY} and F/F_{MSY} computed over each season
- $P(Kobe = green)$ computed across all seasons and iter per year
- SB_0

Experimental design

- Reference OM (R2b): LL CPUE 1 (NW), SSB prior, recruitment variance, F penalty.
- Robustness OMs (R3b): 1% effort creep.
- Robustness tests
  - Low vs. high future recruitment regimes
  - Faster growth, and maturity, lower max size
  - Larger recruitment variability
  - CPUE precision & bias, hyper-stability
  - Implementation error (?)
- Tuning for 50, 60 and 70% P(Kobe=green)
- CPUE LL1 + buffer HCR MP
- JABBA (LL1) + buffer HCR

Current status

- OM grid finalized
  - Update OMs to 2023, project catch
- Both MPs coded and running
- Bug in projection code, review ongoing
  - Only asparent while running MPs
  - Initial years target catch not achieved

Timeline

- Demonstration set tuned MPs and robustness runs
  - Not ready for TCMP deadline
  - Extra time available?
- Presentation to WPMTmT, comparison with assessment
- Updated complete analysis to WPM